PWM TO SPEED CHARACTERISTIC OF A DC MOTOR

1.0 Learning Outcomes.....1
2.0 Hardware Set-Up.....1
3.0 Practical Investigation via Simulink.....1
4.0 Concluding Remarks.....3
Recommended Citation.....4

1.0 Learning Outcomes

Upon completing this practical exercise, students should be able to demonstrate the Learning Outcomes presented in Table 1. At the end of the exercise, revisit each Learning Outcome and confirm that it has been achieved.

Table 1: *Learning Outcomes for the Practical Exercise*

ID	Learning Outcome	Achieved
1	Apply a staircase of PWM values to the DC motor and measure the steady-state speed at each level.	☐
2	Plot the static characteristic of the motor in both directions and identify the dead-zone and any saturation.	☐
3	Judge over which range of PWM values a linear model of the motor is reasonable, and explain the effect of the dead-zone on a controller.	☐

2.0 Hardware Set-Up

This practical exercise builds on the hardware and algorithm set-up described in the practical exercise, DC Motor Speed Measurement using an Encoder, which details the operation of a DC motor with an encoder and the measurement of motor speed in revolutions per minute (RPM). The set-up is not repeated here.

The practical exercise, DC Motor Polarity Control, is also used, so that the motor can be driven in both directions by changing the polarity of the H-bridge.

3.0 Practical Investigation via Simulink

MATLAB and Simulink should be used to complete this practical exercise. A slow staircase of PWM values is applied to the motor, and the steady speed at each step is extracted to give the static characteristic: the steady-state relationship between the PWM value and the speed. The exercise supports the ACE-CORE learning guide, DC Motor Actuators.

Background: The Static Characteristic

An ideal DC motor driven by an ideal H-bridge has a speed proportional to the applied voltage. In practice, friction stops the motor from turning until the voltage is high enough to overcome it, which produces a dead-zone. The voltage at which

the motor starts is usually higher than the voltage at which it stops, because static friction is larger than running friction. At high PWM values, voltage losses in the H-bridge and the supply can also flatten the characteristic.

Apply a PWM Staircase

The staircase increases the PWM value by 15 every 2 s, from 0 to 255. A MATLAB Function block driven by a Clock generates it, see Figure 1.

MATLAB Function Block: generating a PWM staircase from the simulation time

```
function pwm = fcn(t)
% Steps of 15 every 2 s from 0 up to 255 (18 levels)
pwm = min(255, 15*floor(t/2));
```

[Screenshot to insert] *Clock → MATLAB Function (staircase) → PWM output and direction pin; encoder speed (RPM) → Scope. Scopes log the PWM value, speed and time.*

Figure 1: *Simulink Model for Measuring the Static Characteristic*

1. Build the model in Figure 1 with a Stop Time of 36 s, and log the PWM value, the measured speed and the time to the workspace as *u*, *y* and *t*.
2. Run the staircase in the forward direction.
3. Reverse the polarity of the motor and run the staircase again.
4. Change the function to `pwm = max(0, 255 - 15*floor(t/2));` and run a downward staircase in the forward direction.
5. For each run, extract the steady speed at each level with the script below, and record the starting and stopping PWM values in Table 2.

MATLAB Script: extracting the steady speed at each PWM level

```
u_k = u.signals.values(:);  y_k = y.signals.values(:);  t_k =
t.signals.values(:);
levels = unique(u_k);  w_ss = zeros(size(levels));
for i = 1:length(levels)
    idx  = find(u_k == levels(i));
    last = idx(t_k(idx) > t_k(idx(end)) - 1);   % last 1 s at this level
    w_ss(i) = mean(y_k(last));
end
plot(levels, w_ss, 'o-'); grid on;
xlabel('PWM value'); ylabel('Steady-state speed [RPM]');
```

An illustrative characteristic is shown in Figure 2. It is based on the A1.4 example gain of 28 RPM/V with an assumed dead-zone, so the dead-zone width and the gap between the two curves will differ for your motor. Above the dead-zone the speed rises almost linearly with the PWM value, and the motor stops at a lower PWM value on the way down than it starts at on the way up.

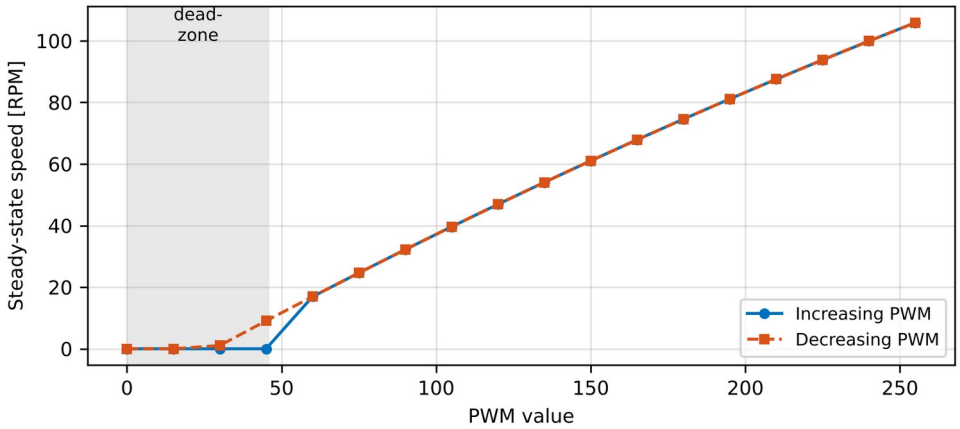


Figure 2: *Illustrative Static Characteristic with an Assumed Dead-Zone (Not Measured Data)*

Table 2: *Dead-Zone and Linear Range for Your Motor*

Run	PWM at which motor starts	PWM at which motor stops	Speed at PWM 255 (RPM)	Linear range (PWM)
Forward, increasing		–		
Reverse, increasing		–		
Forward, decreasing	–			

Interpret the Characteristic

1. Over what range of PWM values is a straight line a reasonable fit to the characteristic? Fit a line to that range with `polyfit` and compare its slope with the gain from the practical exercise, Parameter Estimation of a DC Motor using Linear Least Squares.
2. Explain how the dead-zone would affect a proportional controller trying to hold a low speed.
3. A simple way to compensate for the dead-zone is to add the starting PWM value to the controller output whenever it is not zero. Describe one problem this could cause.

4.0 Concluding Remarks

This practical exercise measured the static characteristic of a DC motor, the steady-state relationship between the PWM value and the speed. Using MATLAB, Simulink and ACE-Lab hardware, a staircase input was applied in both directions and the steady speed at each level was extracted.

The characteristic showed a dead-zone at low PWM values, a different starting and stopping point, and a range in which the speed is close to proportional to the PWM value. A linear model is only valid inside that range, which is why controllers

are designed around an operating point and why the dead-zone matters most when controlling low speeds.

Recommended Citation

If you have used this practical exercise in your teaching, learning or research, please cite it using the reference below, adapting the format as required by your preferred referencing style.

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